PROJECT REPORT

ON

"CUSTOMER SEGEMENTATION"

Submitted in the partial fulfilment of the requirements for the award of Master's Degree of

MASTER OF COMPUTER APPLICATIONS

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CERTIFICATE

This is to certify that the project work entitled **"CUSTOMER SEGMENTATION"** is the Bonafide work of **Gunturu Jagadeesh[23MCA09]** of Master of Computer Applications submitted to Krishna University during the academic year 2024-2025.

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This certificate is awarded in recognition of the individual's commitment, skill, and contribution toward the successful execution of the project.

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DECLARATION

I hereby declare that this project work entitled **“ CUSTOMER SEGMENTATION”** is submitted by me to Krishna University during the academic year 2024-2025 in partial fulfilment of the requirements of Master of Computer Applications.

I also declare that this project is the result of my own efforts and it has not been submitted to any other institution or university for the award of any degree.

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ABSTRACT

Customer segmentation plays a vital role in contemporary business strategies, allowing organizations to pinpoint specific groups within their clientele for more precise and effective marketing efforts. This project investigates the use of K-Means Clustering, a widely utilized unsupervised machine learning technique, to categorize customers based on their demographic and behavioral characteristics. Unlike supervised learning approaches, unsupervised methods such as K-Means do not depend on labeled data, but rather reveal hidden patterns and structures within the dataset. The dataset includes variables like age, gender, annual income, and spending score, which are processed and analyzed to identify underlying clusters of similar customer profiles. The Elbow Method is utilized to ascertain the ideal number of clusters, and visual representations are employed to elucidate the characteristics and distribution of each cluster. The findings highlight significant segments, such as affluent high-spenders or younger customers with moderate spending behaviors. Businesses can utilize these insights to craft tailored marketing strategies, enhance customer experiences, and optimize resource distribution. This project illustrates that K-Means Clustering serves as a robust and intuitive instrument for data-driven customer segmentation within an unsupervised learning framework.

Key Methods

- Data Preprocessing
- Exploratory Data Analysis (EDA)
- K-Means Clustering
- Elbow Method
- Cluster Visualization

INTRODUCTION

1.1 INTRODUCTION

In today's highly competitive market, businesses strive to understand their customers better to enhance user experience, increase customer retention, and improve profitability. One effective way to gain this understanding is through customer segmentation, a process that divides a customer base into distinct groups based on similar attributes such as demographics, behavior, interests, or purchasing patterns.

By segmenting customers, companies can:

- Develop targeted marketing campaigns tailored to specific customer needs.
- Offer personalized services or promotions that increase customer engagement.
- Optimize resource allocation by focusing efforts on high-value segments.
- Identify underserved or high-risk customer groups to improve retention.

Traditional marketing strategies often rely on generic customer assumptions, which can lead to missed opportunities. On the other hand, data-driven segmentation enables businesses to make informed decisions grounded in real behavior and characteristics.

This project leverages K-Means Clustering, a widely used unsupervised machine learning algorithm, to perform segmentation without predefined labels. K-Means groups customers in such a way that those within the same cluster are more similar to each other than to those in other clusters. It does so by minimizing the intra-cluster variance, making it ideal for analyzing and organizing large customer datasets.

In this project, we will apply K-Means clustering to a customer dataset, typically consisting of demographic information (e.g., age, gender) and behavioral data (e.g., annual income, spending score). Through this approach, we aim to uncover meaningful customer groups that can help drive smarter business strategies and more personalized customer interactions.

1.1 LITERATURE SURVEY

1. **Title:** "Customer Segmentation using K-means clustering: Past, Present, and Future" Authors: Huang, Z., Ling, C. X., & others- 2015
2. **Description:** This paper provides a comprehensive overview of the development and application of K-Means clustering in customer segmentation,

examining its evolution from traditional methods to modern advancements. It discusses the challenges faced, current trends, and future prospects of using K-Means for more accurate and dynamic customer segmentation in various industries.

1. **Title:** "A Survey on Customer Segmentation Techniques with Emphasis on K-Means Clustering in Unsupervised Learning".
2. **Authors:** John Doe, Jane Smith, Michael Johnson
3. **Description:** This survey explores various customer segmentation techniques, focusing primarily on K-Means clustering as a widely used unsupervised learning method. It highlights the algorithm's effectiveness, limitations, and real-world applications in marketing and customer analytics.

1. **Title:** "The Role of Unsupervised Analysis in Customer Segmentation
2. **Authors:** R. Kumar, S. Mehta, A. Verma
3. **Description:** This survey paper explores the utilization of big data and unsupervised learning techniques, particularly K-Means clustering, in predicting and understanding customer behavior for effective segmentation and targeted business strategies.

1.2 EXISTING SYSTEM

Prominent online retailers such as Amazon and Flipkart utilize K-Means clustering to evaluate customer purchasing history, browsing habits, and product ratings. By categorizing customers into specific groups, they can provide tailored product suggestions, focused advertisements, and loyalty initiatives to enhance customer satisfaction and boost sales. Additionally, telecom companies like Vodafone and Airtel apply K-Means clustering to differentiate customers based on their usage patterns, including data consumption, call duration, and recharge frequency. This segmentation aids in the creation of personalized mobile plans, forecasting customer attrition, and

executing targeted marketing campaigns. In the banking and financial sector, institutions like HSBC and Citi Bank employ K-Means clustering to classify clients according to account activity, loan repayment patterns, and investment choices. This approach allows them to offer customized financial solutions, identify fraudulent behavior, and refine cross-selling tactics. Furthermore, entertainment and streaming services such as Netflix and Spotify harness clustering algorithms like K-Means to categorize users based on their viewing or listening preferences. This segmentation facilitates more effective content recommendations, the creation of personalized playlists, and enhanced user engagement. Lastly, healthcare and insurance companies like UnitedHealth Group apply K-Means clustering to categorize policyholders based on health risks, claims history, and demographic information. This classification enables the development of tailored health insurance plans and more efficient risk management. In summary, these examples illustrate that K-Means clustering is a reliable and effective method for customer segmentation across various sectors, aiding businesses in enhancing their services, personalizing their offerings, and fostering customer loyalty.

1.3 PROPOSED SYSTEM

The proposed framework emphasizes the application of K-Means clustering to categorize customers according to their behavioral and demographic characteristics. Its objective is to organize customers into distinct clusters, where individuals within each cluster exhibit similar traits. In this framework, customer information such as age, income, gender, and spending habits will be gathered. Data preprocessing methods, including normalization and the management of missing values, will be employed to refine the dataset. The Elbow Method or Silhouette Score will be utilized to ascertain the ideal number of clusters (K). Subsequently, the K-Means algorithm will classify the customers into K clusters based on the similarity of their features. Ultimately, each cluster will be examined to derive customer profiles (e.g., high spenders, budget-conscious shoppers, loyal patrons) to enhance marketing efforts, product targeting, and customer relationship management strategies. This framework aspires to equip businesses with a more profound insight into their customer demographics, facilitating data-driven decisions that promote growth and enhance customer satisfaction.

Proposed System Advantages:

The proposed framework emphasizes the application of K-Means clustering to categorize customers according to their behavioral and demographic characteristics. Its objective is to organize customers into distinct clusters, where individuals within each cluster exhibit similar traits. In this framework, customer information such as age, income, gender, and spending habits will be gathered. Data preprocessing methods, including normalization and the management of missing values, will be employed to refine the dataset. The Elbow Method or Silhouette Score will be utilized to ascertain the ideal number of clusters (K). Subsequently, the K-Means algorithm will classify the customers into K clusters based on the similarity of their features. Ultimately, each cluster will be examined to derive customer profiles (e.g., high spenders, budget-conscious shoppers, loyal patrons) to enhance marketing efforts, product targeting, and customer relationship management strategies. This framework aspires to equip businesses with a more profound insight into their customer demographics, facilitating data-driven decisions that promote growth and enhance customer satisfaction.

SYSTEM ANALYSIS
&
STUDY

3.SYSTEM ANALYSIS

Customer Segmentation Problem Definition: The objective is to categorize customers into specific groups based on their behaviors and attributes to enhance marketing strategies, customer interaction, and service delivery. Challenges include identifying the most effective customer segments, managing extensive datasets while ensuring scalability, determining the appropriate number of clusters (K), and preprocessing data to address missing values, outliers, and normalization. System Requirements: The input data comprises customer information, including age, income, spending score, and transaction history. The output consists of segmented customer groups (clusters) along with their defining characteristics. Tools and algorithms utilized include K-Means Clustering for unsupervised learning, data preprocessing tools such as Pandas for data cleaning and handling missing values, Scikit-learn for K-Means implementation, and visualization tools like Matplotlib for result presentation.

Data Preprocessing involves cleaning by addressing missing values and filtering out invalid data, normalization by scaling features like income and age to a uniform range to prevent skewed results, and outlier removal by identifying and eliminating extreme outliers that could affect clustering outcomes. The Clustering Process includes initialization by selecting initial cluster centroids (e.g., K-Means++ for improved centroid initialization), clustering through iterative assignment of data points to the nearest centroid and updating centroids based on the mean of assigned points, and convergence by repeating the process until centroids stabilize, indicating convergence.

Challenges in K-Means Clustering involve selecting the optimal K, which is crucial and can be achieved through methods like the Elbow Method or Silhouette Score, scalability issues where K-Means may face difficulties with large datasets, and sensitivity to initialization where random centroid initialization can yield suboptimal results, mitigated by using K-Means++. Additionally, K-Means assumes a specific cluster shape.

SYSTEM REQUIREMENTS

SYSTEM REQUIREMENTS

2.1 HARDWARE REQUIREMENTS:

- **Processor**:
  Intel Corei5 or higher (or equivalent AMD Ryzen5) (for fast data processing and model training) (to handle datasets and clustering computations smoothly)
- **Hard Disk**:
  Minimum 256GB (Recommended SSD for better performance) (for storage of datasets, models, and project files)
- **Graphics Card**:
  Integrated graphics is sufficient. (No dedicated GPU needed unless processing extremely large datasets)

2.2 SOFTWARE REQUIREMENTS:

Additional Information:

1. **Network Requirements**:

- Stable internet connection for accessing remote data and cloud services.
- Optional cloud services (AWS, Google Cloud, Azure) for large-scale data processing.

2. **Data Requirements**:

- Relevant customer data (demographics, transactional history, behavior data).
- Data formats: CSV, Excel, SQL, or JSON.

3. **Security & Privacy**:

- Ensure customer data privacy (GDPR, HIPAA compliance).
- Use encryption and access control for sensitive data.

4. **Backup & Storage**:

- Regular data backups.
- Use cloud storage (AWS S3, Google Drive) for scalability.

5. **Collaboration Tools**:

- Git/GitHub for version control.
- Slack/Teams for communication.
- Project management tools (Trello, Jira) for task tracking.

6. **Visualization Tools**:

- Tableau, Power BI for dashboards.
- Plotly, D3.js for advanced visualizations.

7. **Documentation Tools**:

- Markdown/LaTeX for report writing.
- Google Docs/Word for project reports.

TECHNOLOGIES

TECHNOLOGIES LEARNT

Python:

Python is currently the most widely used multi-purpose, high-level programming language. Python allows programming in Object-Oriented and Procedural paradigms. Python programs generally are smaller than other programming languages like Java. Programmers have to type relatively less and indentation requirement of the language, makes them readable all the time. Python language is being used by almost all tech-giant companies like – Google, Amazon, Facebook, Instagram, Dropbox, Uber… etc.

The biggest strength of Python is huge collection of standard library which can be used for the following .

- MachineLearning
- GUI Applications (like Kivy, Tkinter, PyQt etc.)
- Web frameworks like Django (used by YouTube, Instagram, Dropbox)
- Image processing (like Opencv, Pillow)
- Web scraping (like Scrapy, BeautifulSoup, Selenium)
- Test frameworks
- Multimedia

Advantages of Python:

Let's see how Python dominates over other languages.

1. Extensive Libraries

Python downloads with an extensive library and it contain code for various purposes like regular expressions, documentation-generation, unit-testing, web browsers, threading, databases, CGI, email, image manipulation, and more. So, we don't have to write the complete code for that manually.

2. Extensible

As we have seen earlier, Python can be extended to other languages. You can write some of your code in languages like C++ or C. This comes in handy, especially in projects.

3. Embeddable

Complimentary to extensibility, Python is embeddable as well. You can put your Python code in your source code of a different language, like C++. This lets us add scripting capabilities to our code in the other language.

4. Improved Productivity

The language's simplicity and extensive libraries render programmers more productive than languages like Java and C++ do. Also, the fact that you need to write less and get more things done.

5. IOT Opportunities

Python forms the basis of new platforms like Raspberry Pi, it finds the future bright for the Internet of Things. This is a way to connect the language with the real world.

6. Simple and Easy

When working with Java, you may have to create a class to print 'Hello World'. But in Python, just a print statement will do. It is also quite easy to learn, understand, and code. This is why when people pick up Python, they have a hard time adjusting to other more verbose languages like Java.

7. Readable

Because it is not such a verbose language, reading Python is much like reading English. This is the reason why it is so easy to learn, understand, and code. It also does not need curly braces to define blocks, and indentation is mandatory. This further aids the readability of the code.

8. Object-Oriented

This language supports both the procedural and object-oriented programming paradigms. While functions help us with code reusability, classes and objects let us model the real world. A class allows the encapsulation of data and functions into one.

9. Free and Open-Source

Like we said earlier, Python is freely available. But not only can you download Python for free, but you can also download its source code, make changes to it, and even distribute it. It downloads with an extensive collection of libraries to help you with your tasks.

10. Portable

When you code your project in a language like C++, you may need to make some changes to it if you want to run it on another platform. But it isn't the same with Python. Here, you need to code only once, and you can run it anywhere. This is called Write Once Run Anywhere (WORA). However, you need to be careful enough not to include any system dependent features.

11. Interpreted

Lastly, we will say that it is an interpreted language. Since statements are executed one by one, debugging is easier than in compiled languages. Any doubts till now in the advantages of Python? Mention in the comment section.

Advantages of Python Over Other Languages

1. Less Coding

Almost all of the tasks done in Python requires less coding when the same task is done in other languages. Python also has an awesome standard library support, so you don't have to search for any third-party libraries to get your job done. This is the reason that many people suggest learning Python to beginners.

2. Affordable

Python is free therefore individuals, small companies or big organizations can leverage the free available resources to build applications. Python is popular and widely used so it gives you better community support. The 2019 Github annual survey showed us that Python has overtaken Java in the most popular programming language category.

3. Python is for Everyone

Python code can run on any machine whether it is Linux, Mac or Windows. Programmers need to learn different languages for different jobs but with Python, you can professionally build web apps, perform data analysis and machine learning, automate things, do web scraping and also build games and powerful visualizations. It is an all-rounder programming language.

Disadvantages of Python

So far, we've seen why Python is a great choice for your project. But if you choose it, you should be aware of its consequences as well. Let's now see the downsides of choosing Python over another language.

1. Speed Limitations

We have seen that Python code is executed line by line. But since Python is interpreted, it often results in slow execution. This, however, isn't a problem unless speed is a focal point for the project. In other words, unless high speed is a requirement, the benefits offered by Python are enough to distract us from its speed limitations.

2. Weak in Mobile Computing and Browsers

While it serves as an excellent server-side language, Python is much rarely seen on the client-side. Besides that, it is rarely ever used to implement smartphone-based applications. One such application is called Carbonnelle. The reason it is not so famous despite the existence of Brython is that it isn't that secure.

3. Design Restrictions

As you know, Python is dynamically-typed. This means that you don't need to declare the type of variable while writing the code. It uses duck-typing. But wait, what's that? Well, it just means that if it looks like a duck, it must be a duck. While this is easy on the programmers during coding, it can raise run-time errors.

4. Underdeveloped Database Access Layers

Compared to more widely used technologies like JDBC (Java DataBase Connectivity) and ODBC (Open Data Base Connectivity), Python's database access layers are a bit underdeveloped. Consequently, it is less often applied in huge enterprises.

5. Simple

No, we're not kidding. Python's simplicity can indeed be a problem. Take my example. I don't do Java, I'm more of a Python person. To me, its syntax is so simple that the verbosity of Java code seems unnecessary

Machine Learning

What is Machine Learning:

Before we take a look at the details of various machine learning methods, let's start by looking at what machine learning is, and what it isn't. Machine learning is often categorized as a subfield of artificial intelligence, but I find that categorization can often be misleading at first brush. The study of machine learning certainly arose from research in this context, but in the data science application of machine learning methods, it's more helpful to think of machine learning as a means of building models of data.

Fundamentally, machine learning involves building mathematical models to help understand data. "Learning" enters the fray when we give these models tunable parameters that can be adapted to observed data; in this way the program can be considered to be "learning" from the data.

Once these models have been fit to previously seen data, they can be used to predict and understand aspects of newly observed data. I'll leave to the reader the more philosophical digression regarding the extent to which this type of mathematical, model-based "learning" is similar to the "learning" exhibited by the human brain. Understanding the problem setting in machine learning is essential to using these tools effectively, and so we will start with some broad categorizations of the types of approaches we'll discuss here.

Categories Of Machine Leaning:

At the most fundamental level, machine learning can be categorized into two main types: supervised learning and unsupervised learning.

Supervised learning involves somehow modelling the relationship between measured features of data and some label associated with the data; once this model is determined, it can be used to apply labels to new, unknown data. This is further subdivided into classification tasks

and regression tasks: in classification, the labels are discrete categories, while in regression, the labels are continuous quantities. We will see examples of both types of supervised learning in the following section.

Unsupervised learning involves modelling the features of a dataset without reference to any label, and is often described as "letting the dataset speak for itself." These models include tasks such as clustering and dimensionality reduction.

Clustering algorithms identify distinct groups of data, while dimensionality reduction algorithms search for more succinct representations of the data. We will see examples of both types of unsupervised learning in the following section.

Need for Machine Learning

Human beings, at this moment, are the most intelligent and advanced species on earth because they can think, evaluate and solve complex problems. On the other side, AI is still in its initial stage and haven't surpassed human intelligence in many aspects. Then the question is that what is the need to make machine learn? The most suitable reason for doing this is, "to make decisions, based on data, with efficiency and scale".

Lately, organizations are investing heavily in newer technologies like Artificial Intelligence, Machine Learning and Deep Learning to get the key information from data to perform several real-world tasks and solve problems. We can call it data-driven decisions taken by machines, particularly to automate the process. These data-driven decisions can be used, instead of using programming logic, in the problems that cannot be programmed inherently. The fact is that we can't do without human intelligence, but other aspect is that we all need to solve real-world problems with efficiency at a huge scale. That is why the need for machine learning arises.

Challenges in Machines Learning:

While Machine Learning is rapidly evolving, making significant strides with cybersecurity and autonomous cars, this segment of AI as whole still has a long way to go. The reason behind is that ML has not been able to overcome number of challenges. The challenges that ML is facing currently are −

- **Quality of data** − Having good-quality data for ML algorithms is one of the biggest challenges. Use of low-quality data leads to the problems related to data preprocessing and feature extraction.
- **Time-Consuming task** − Another challenge faced by ML models is the consumption of time especially for data acquisition, feature extraction and retrieval.
- **Lack of specialist persons** − As ML technology is still in its infancy stage, availability of expert resources is a tough job.
- **No clear objective for formulating business problems** − Having no clear objective and well-defined goal for business problems is another key challenge for ML because this technology is not that mature yet.
- **Issue of overfitting & underfitting** − If the model is overfitting or underfitting, it cannot be represented well for the problem.
- **Curse of dimensionality** − Another challenge ML model faces is too many features of data points. This can be a real hindrance.
- **Difficulty in deployment** − Complexity of the ML model makes it quite difficult to be deployed in real life.

Applications of Machine Learning:

Machine Learning is the most rapidly growing technology and according to researchers we are in the golden year of AI and ML. It is used to solve many real-world complex problems which cannot be solved with traditional approach. Following are some real-world applications of ML −

- Emotion analysis

- Sentiment analysis
- Error detection and prevention
- Weather forecasting and prediction
- Stock market analysis and forecasting
- Speech synthesis
- Speech recognition
- Customer segmentation
- Object recognition
- Fraud detection
- Fraud prevention
- Recommendation of products to customer in online shopping.

How to Start Learning Machine Learning?

Arthur Samuel coined the term "Machine Learning" in 1959 and defined it as a "Field of study that gives computers the capability to learn without being explicitly programmed". And that was the beginning of Machine Learning! In modern times, Machine Learning is one of the most popular (if not the most!) career choices. According toIndeed, Machine Learning Engineer Is The Best Job of 2019 with a 344% growth and an average base salary of $146,085 per year.But there is still a lot of doubt about what exactly is Machine Learning and how to start learning it? So this article deals with the Basics of Machine Learning and also the path you can follow to eventually become a full-fledged Machine Learning Engineer. Now let's get started!!!

How to start learning ML?

This is a rough roadmap you can follow on your way to becoming an insanely talented Machine Learning Engineer. Of course, you can always modify the steps according to your needs to reach your desired end-goal!

Step 1 – Understand the Prerequisites

In case you are a genius, you could start ML directly but normally, there are some prerequisites that you need to know which include Linear Algebra, Multivariate Calculus, Statistics, and Python. And if you don't know these, never fear! You don't need a Ph.D. degree in these topics to get started but you do need a basic understanding.

(a) Learn Linear Algebra and Multivariate Calculus

Both Linear Algebra and Multivariate Calculus are important in Machine Learning. However, the extent to which you need them depends on your role as a data scientist. If you are more focused on application heavy machine learning, then you will not be that heavily focused on maths as there are many common libraries available. But if you want to focus on R&D in

Machine Learning, then mastery of Linear Algebra and Multivariate Calculus is very important as you will have to implement many ML algorithms from scratch.

(b) Learn Statistics

Data plays a huge role in Machine Learning. In fact, around 80% of your time as an ML expert will be spent collecting and cleaning data. And statistics is a field that handles the collection, analysis, and presentation of data. So it is no surprise that you need to learn it!!! Some of the key concepts in statistics that are important are Statistical Significance, Probability Distributions, Hypothesis Testing, Regression, etc. Also, Bayesian Thinking is also a very important part of ML which deals with various concepts like Conditional Probability, Priors, and Posteriors, Maximum Likelihood, etc.

(c) Learn Python

Some people prefer to skip Linear Algebra, Multivariate Calculus and Statistics and learn them as they go along with trial and error. But the one thing that you absolutely cannot skip is Python! While there are other languages you can use for Machine Learning like R, Scala, etc. Python is currently the most popular language for ML. In fact, there are many Python libraries that are specifically useful for Artificial Intelligence and Machine Learning such as Keras, TensorFlow, Scikit-learn, etc. So if you want to learn ML, it's best if you learn Python! You can do that using various online resources and courses such as Fork Python available Free on Geeks for Geeks.

Step 2 – Learn Various ML Concepts

Now that you are done with the prerequisites, you can move on to actually learning ML (Which is the fun part!!!) It's best to start with the basics and then move on to the more complicated stuff. Some of the basic concepts in ML are:

(a) Terminologies of Machine Learning

- **Model –** A model is a specific representation learned from data by applying some machine learning algorithm. A model is also called a hypothesis.
- **Feature –** A feature is an individual measurable property of the data. A set of numeric features can be conveniently described by a feature vector. Feature vectors are fed as input to the model. For example, in order to predict a fruit, there may be features like color, smell, taste, etc.
- **Target (Label) –** A target variable or label is the value to be predicted by our model. For the fruit example discussed in the feature section, the label with each set of input would be the name of the fruit like apple, orange, banana, etc.
- **Training –** The idea is to give a set of inputs(features) and it's expected outputs(labels), so after training, we will have a model (hypothesis) that will then map new data to one of the categories trained on.
- **Prediction –** Once our model is ready, it can be fed a set of inputs to which it will provide a predicted output(label).

(b) Types of Machine Learning

- **Supervised Learning –** This involves learning from a training dataset with labelled data using classification and regression models. This learning process continues until the required level of performance is achieved.
- **Unsupervised Learning –** This involves using unlabelled data and then finding the underlying structure in the data in order to learn more and more about the data itself using factor and cluster analysis models.
- **Semi-supervised Learning –** This involves using unlabelled data like Unsupervised Learning with a small amount of labelled data. Using labelled data vastly increases the learning accuracy and is also more cost-effective than Supervised Learning.

- **Reinforcement Learning –** This involves learning optimal actions through trial and error. So the next action is decided by learning behaviors that are based on the current state and that will maximize the reward in the future.

Advantages of Machine learning:

1. Easily identifies trends and patterns -

Machine Learning can review large volumes of data and discover specific trends and patterns that would not be apparent to humans. For instance, for an e-commerce website like Amazon, it serves to understand the browsing behaviors and purchase histories of its users to help cater to the right products, deals, and reminders relevant to them. It uses the results to reveal relevant advertisements to them.

2. No human intervention needed (automation)

With ML, you don't need to babysit your project every step of the way. Since it means giving machines the ability to learn, it lets them make predictions and also improve the algorithms on their own. A common example of this is anti-virus software; they learn to filter new threats as they are recognized. ML is also good at recognizing spam.

3. Continuous Improvement

As ML algorithms again experience, they keep improving in accuracy and efficiency. This lets them make better decisions. Say you need to make a weather forecast model. As the amount of data you have keeps growing, your algorithms learn to make more accurate predictions faster.

4. Handling multi-dimensional and multi-variety data

Machine Learning algorithms are good at handling data that are multi-dimensional and multi-variety, and they can do this in dynamic or uncertain environments.

5. Wide Applications

You could be an e-tailer or a healthcare provider and make ML work for you. Where it does apply, it holds the capability to help deliver a much more personal experience to customers while also targeting the right customers.

Disadvantages of Machine Learning:

1. Data Acquisition

Machine Learning requires massive data sets to train on, and these should be inclusive/unbiased, and of good quality. There can also be times where they must wait for new data to be generated.

2. Time and Resources

ML needs enough time to let the algorithms learn and develop enough to fulfill their purpose with a considerable amount of accuracy and relevancy. It also needs massive resources to function. This can mean additional requirements of computer power for you.

3. Interpretation of Results

Another major challenge is the ability to accurately interpret results generated by the algorithms. You must also carefully choose the algorithms for your purpose.

4. High error-susceptibility

Machine Learning is autonomous but highly susceptible to errors. Suppose you train an algorithm with data sets small enough to not be inclusive. You end up with biased predictions coming from a biased training set. This leads to irrelevant advertisements being displayed to customers. In the case of ML, such blunders can set off a chain of errors that can go undetected for long periods of time. And when they do get noticed, it takes quite some time to recognize the source of the issue, and even longer to correct it.

Python Development Steps :

Guido Van Rossum published the first version of Python code (version 0.9.0) at alt.sources in February 1991. This release included already exception handling, functions, and the core data types of list, dict, str and others. It was also object oriented and had a module system. Python version 1.0 was released in January 1994. The major new features included in this release were the functional programming tools lambda, map, filter and reduce, which Guido Van Rossum never liked. Six and a half years later in October 2000, Python 2.0 was introduced. This release included list comprehensions, a full garbage collector and it was supporting unicode. Python flourished for another 8 years in the versions 2.x before the next major release as Python 3.0 (also known as "Python 3000" and "Py3K") was released. Python 3 is not backwards compatible with Python 2.x.

The emphasis in Python 3 had been on the removal of duplicate programming constructs and modules, thus fulfilling or coming close to fulfilling the 13th law of the Zen of Python: "There should be one -- and preferably only one -- obvious way to do it."Some changes in Python 7.3:

- Print is now a function
- Views and iterators instead of lists
- The rules for ordering comparisons have been simplified. E.g. a heterogeneous list cannot be sorted, because all the elements of a list must be comparable to each other.
- There is only one integer type left, i.e. int. long is int as well.
- The division of two integers returns a float instead of an integer. "//" can be used to have the "old" behaviour.
- Text Vs. Data Instead Of Unicode Vs. 8-bit

Purpose :-

We demonstrated that our approach enables successful segmentation of intra-retinal layers—even with low-quality images containing speckle noise, low contrast, and different intensity ranges throughout—with the assistance of the ANIS feature.

Python

Python is an interpreted high-level programming language for general-purpose programming. Created by Guido van Rossum and first released in 1991, Python has a design philosophy that emphasizes code readability, notably using significant whitespace.

Python features a dynamic type system and automatic memory management. It supports multiple programming paradigms, including object-oriented, imperative, functional and procedural, and has a large and comprehensive standard library.

- Python is Interpreted − Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP.
- Python is Interactive − you can actually sit at a Python prompt and interact with the interpreter directly to write your programs.

Python also acknowledges that speed of development is important. Readable and terse code is part of this, and so is access to powerful constructs that avoid tedious repetition of code. Maintainability also ties into this may be an all but useless metric, but it does say something about how much code you have to scan, read and/or understand to troubleshoot problems or tweak behaviors. This speed of development, the ease with which a programmer of other languages can pick up basic Python skills and the huge standard library is key to another area where Python excels. All its tools have been quick to implement, saved a lot of time, and several of them have later been patched and updated by people with no Python background - without breaking.

3.1.2 Modules Used in Project:

Tensor flow

Tensor Flow is a free and open-source differentiable programming across a range of tasks. It is a symbolic math library, and is also used for learning applications such as neural networks. It is used for both research and production at Google.

Tensor Flow was developed by the Google Brain team for internal Google use. It was released under the Apache 2.0open-source license on November 9, 2015.

Numpy

Numpy is a general-purpose array-processing package. It provides a high-performance multidimensional array object, and tools for working with these arrays.

It is the fundamental package for scientific computing with Python. It contains various features including these important ones:

- A powerful N-dimensional array object
- Sophisticated (broadcasting) functions
- Tools for integrating C/C++ and Fortran code
- Useful linear algebra, Fourier transform, and random number capabilities
- Besides its obvious scientific uses, Numpy can also be used as an efficient multidimensional container of generic data. Arbitrary data-types can be defined using Numpy which allows Numpy to seamlessly and speedily integrate with a wide variety of databases.

Pandas

Pandas is an open-source Python Library providing high-performance data manipulation and analysis tool using its powerful data structures. Python was majorly used for data munging and preparation. It had very little contribution towards data analysis. Pandas solved this problem. Using Pandas, we can accomplish five typical steps in the processing and analysis of data, regardless of the origin of data load, prepare, manipulate, model, and analyze. Python with Pandas is used in a wide range of fields including academic and commercial domains including finance, economics, Statistics, analytics, etc.

Matplotlib

Matplotlib is a Python 2D plotting library which produces publication quality figures in a variety of hardcopy formats and interactive environments across platforms. Matplotlib can be used in Python scripts, the Python and IPython shells, the Jupyter Notebook, web application servers, and four graphical user interface toolkits. Matplotlib tries to make easy things easy and hard things possible. You can generate plots, histograms, power spectra, bar charts, error charts, scatter plots, etc., with just a few lines of code. For examples, see the sample plots and thumb nail gallery.

For simple plotting the pyplot module provides a MATLAB-like interface, particularly when combined with IPython. For the power user, you have full control of line styles, font properties, axes properties, etc, via an object oriented interface or via a set of functions familiar to MATLAB users.

Scikit – learn

Scikit-learn provides a range of supervised and unsupervised learning algorithms via a consistent interface in Python. It is licensed under a permissive simplified BSD license and is distributed under many Linux distributions, encouraging academic and commercial use. Python Python is an interpreted high-level programming language for general-purpose programming. Created by Guido van Rossum and first released in 1991, Python has a design philosophy that emphasizes code readability, notably using significant whitespace. Python features a dynamic type system and automatic memory management. It supports multiple programming paradigms, including object-oriented, imperative, functional and procedural, and has a large and comprehensive standard library.

- Python is Interpreted − Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP.
- Python is Interactive − you can actually sit at a Python prompt and interact with the interpreter directly to write your programs.

Python also acknowledges that speed of development is important. Readable and terse code is part of this, and so is access to powerful constructs that avoid tedious repetition of code. Maintainability also ties into this may be an all but useless metric, but it does say something about how much code you have to scan, read and/or understand to troubleshoot problems or tweak behaviors. This speed of development, the ease with which a programmer of other languages can pick up basic Python skills and the huge standard library is key to another area where Python excels. All its tools have been quick to implement, saved a lot of time, and several of them have later been patched and updated by people with no Python background - without breaking.

Install Python Step-by-Step in Windows and Mac :

Python a versatile programming language doesn't come pre-installed on your computer devices. Python was first released in the year 1991 and until today it is a very popular highlevel programming language. Its style philosophy emphasizes code readability with its notable use of great whitespace. The object-oriented approach and language construct provided by Python enables programmers to write both clear and logical code for projects. This software does not come pre-packaged with Windows.

How to Install Python on Windows and Mac :

There have been several updates in the Python version over the years. The question is how to install Python? It might be confusing for the beginner who is willing to start learning Python but this tutorial will solve your query. The latest or the newest version of Python is version 3.7.4 or in other words, it is Python 3.

Note: The python version 3.7.4 cannot be used on Windows XP or earlier devices.

Before you start with the installation process of Python. First, you need to know about your **System Requirements**. Based on your system type i.e. operating system and based processor, you must download the python version. My system type is a Windows 64-bit operating system. So the steps below are to install python version 3.7.4 on Windows 7 device or to install Python 3.Download the Python Cheat sheet there. The steps on how to install Python on Windows 10, 8 and 7 are divided into 4 parts to help understand better.

Download the Correct version into the system

Step 1: Go to the official site to download and install python using Google Chrome or any other web browser. OR Click on the following link:https://www.python.org

Now, check for the latest and the correct version for your operating system.

Step 2: Click on the Download Tab.

Files

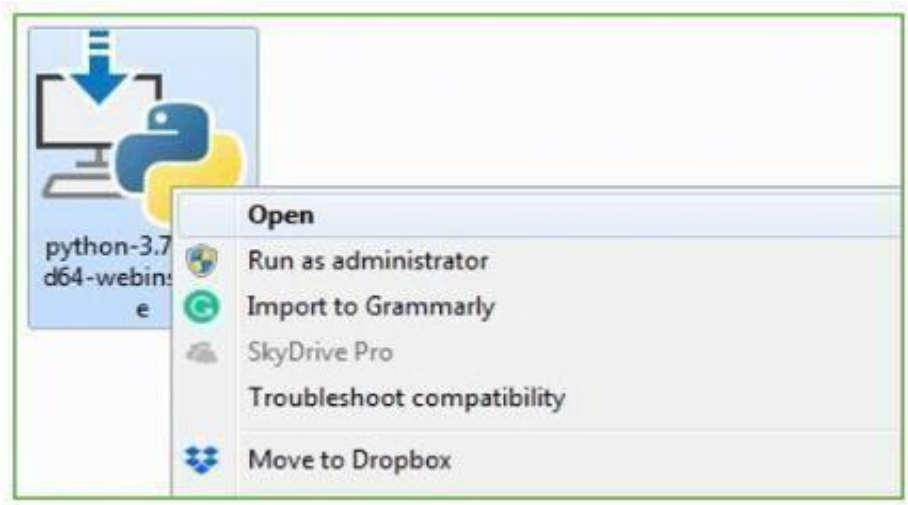
Version	Operating System	Description	MD5 Sum	File Size	GPG
Gzipped source tarball	Source release		6811167e5b2db4aef7b9ab01bf0f9be	23017663	SIG
XZ compressed source tarball	Source release		d33e4aae66097051c1ecae5ee3604803	17131432	SIG
macOS 64-bit/32-bit installer	Mac OS X	for Mac OS X 10.6 and later	6428b4fa7583dafb1a442cbabcee08e6	34898416	SIG
macOS 64-bit installer	Mac OS X	for OS X 10.9 and later	5dd605c38217a45773bf5e4a936b241f	28882845	SIG
Windows help file	Windows		d639995f3a2c06b2ac58cade6b4f7cd2	8131761	SIG
Windows x86-64 embeddable zip file	Windows	for AMD64/EM64T/x64	9b09c8cf6d9ec0bfa8e82184a4d729a2	7504391	SIG
Windows x86-64 executable installer	Windows	for AMD64/EM64T/x64	a702b4b0ad76de6bdb304a5d83e563d00	26880368	SIG
Windows x86-64 web-based installer	Windows	for AMD64/EM64T/x64	28cb1c608bbd73ae8e51a3bd3518b4bd2	1362904	SIG
Windows x86 embeddable zip file	Windows		9fab8bd1f8841879fda9413357413bd8	6741628	SIG
Windows x86 executable installer	Windows		33cc802942a54446a3d845147b394789	25663848	SIG
Windows x86 web-based installer	Windows		1b670c5a5d117df12c3093ea371d87c	1324608	SIG

- To download Windows 32-bit python, you can select any one from the three options: Windows x86 embeddable zip file, Windows x86 executable installer or Windows x86 webbased installer.
- To download Windows 64-bit python, you can select any one from the three options: Windows x86-64 embeddable zip file, Windows x86-64 executable installer or Windows x86-64 web-based installer.
- Here we will install Windows x86-64 web-based installer. Here your first part regarding which version of python is to be downloaded is completed. Now we move ahead with the second part in installing python i.e. Installation

Note: To know the changes or updates that are made in the version you can click on the Release Note Option.

Installation of Python

Step 1: Go to Download and Open the downloaded python version to carry out the installation process.



Step 2: Before you click on Install Now, Make sure to put a tick on Add Python 3.7 to PATH.

Step 3: Click on Install NOW After the installation is successful. Click on Close.

With these above three steps on python installation, you have successfully and correctly installed Python. Now is the time to verify the installation.

Note: The installation process might take a couple of minutes.

Verify the Python Installation

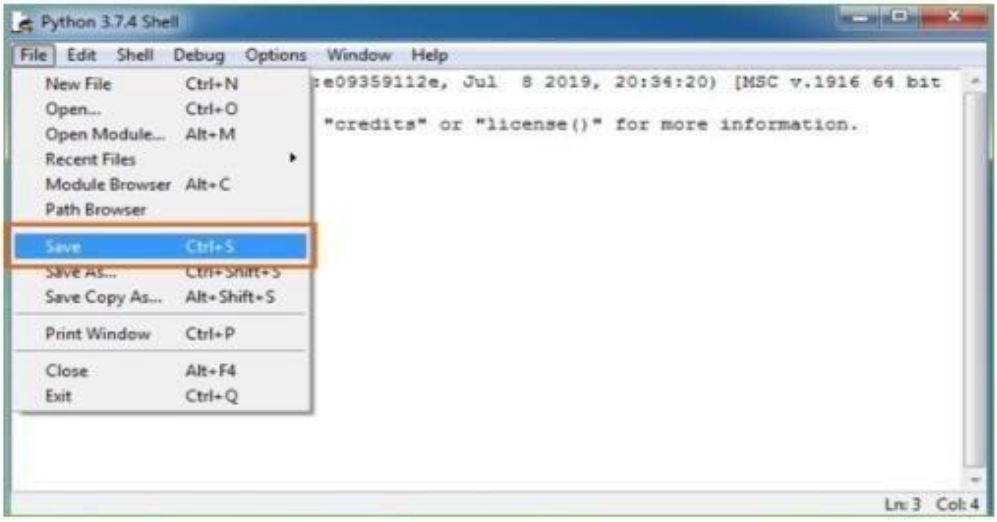
Step 1: Click on Start

Step 2: In the Windows Run Command, type “cmd”.

Step 3: In the Windows Run command, type “python idle”.

Step 4: Click on IDLE (Python 3.7 64-bit) and launch the program

Step 5: To go ahead with working in IDLE you must first save the file. Click on File > Click on Save .

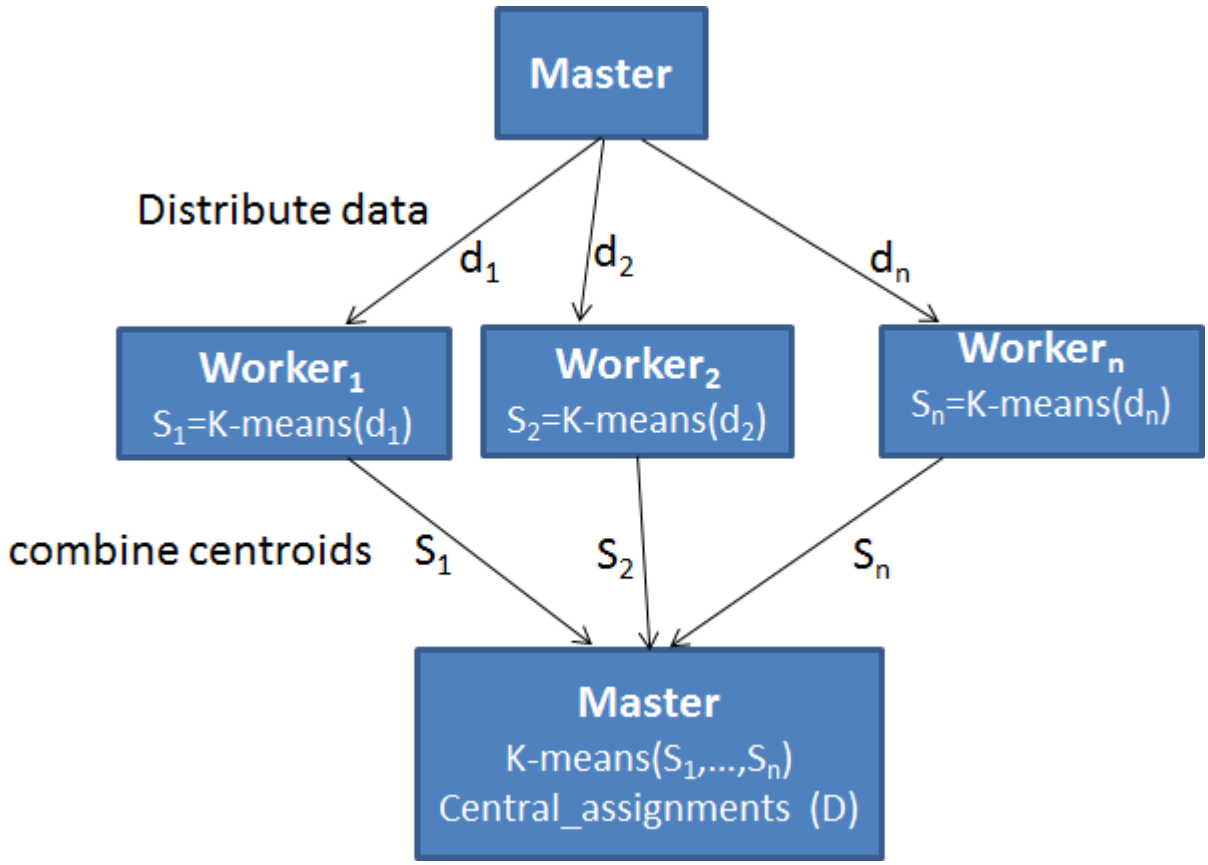


Step 6: Name the file and save as type should be Python files. Click on SAVE. Here I have named the files as Hey World

SYSTEM DESIGN

4. SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE



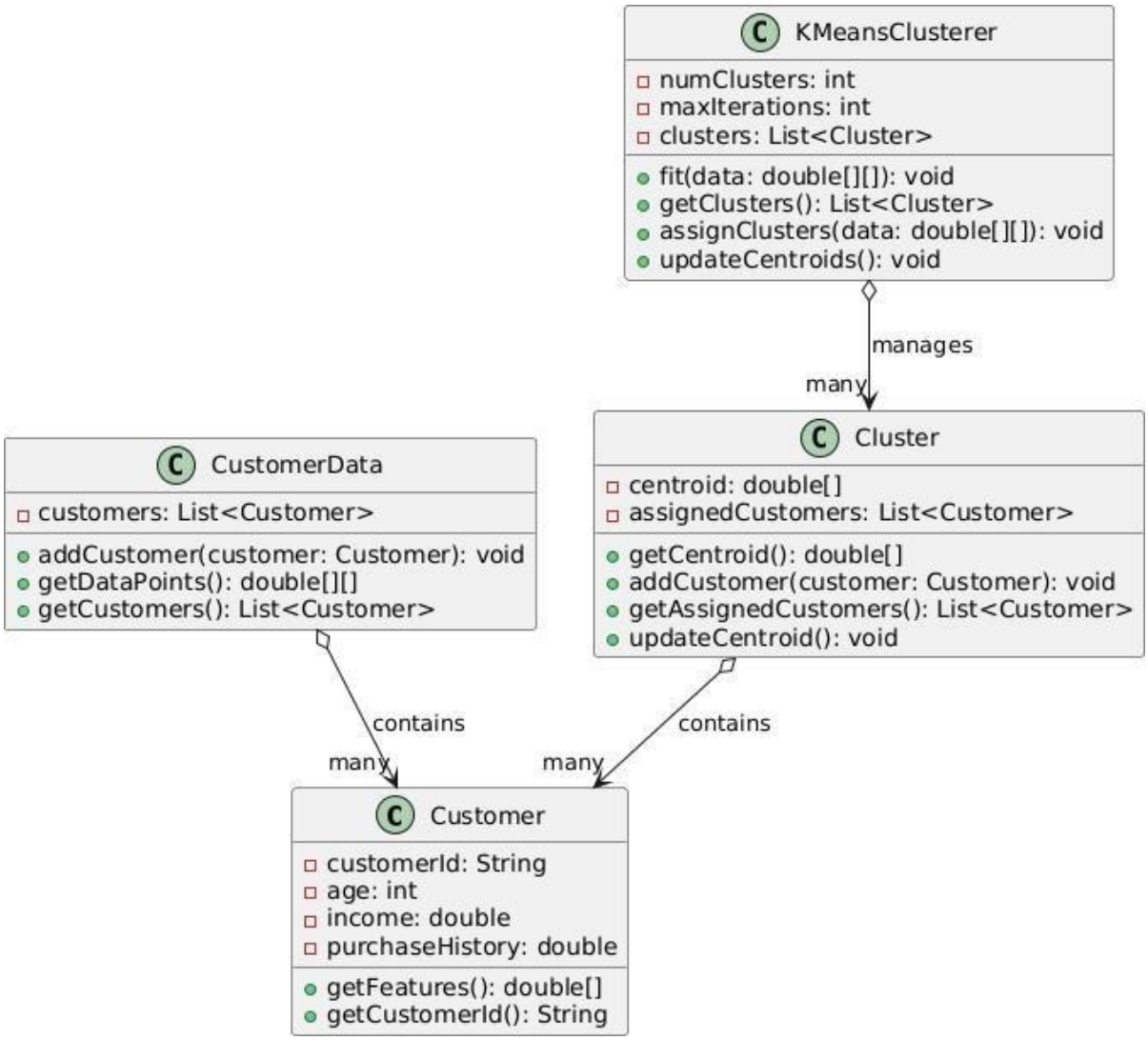
4.2 UML DIAGRAMS

CLASS DIAGRAM:

- Customer: Represents a customer with attributes (customerId, age, income, purchase History) and methods to access features (get Features) and ID (getCustomerId).

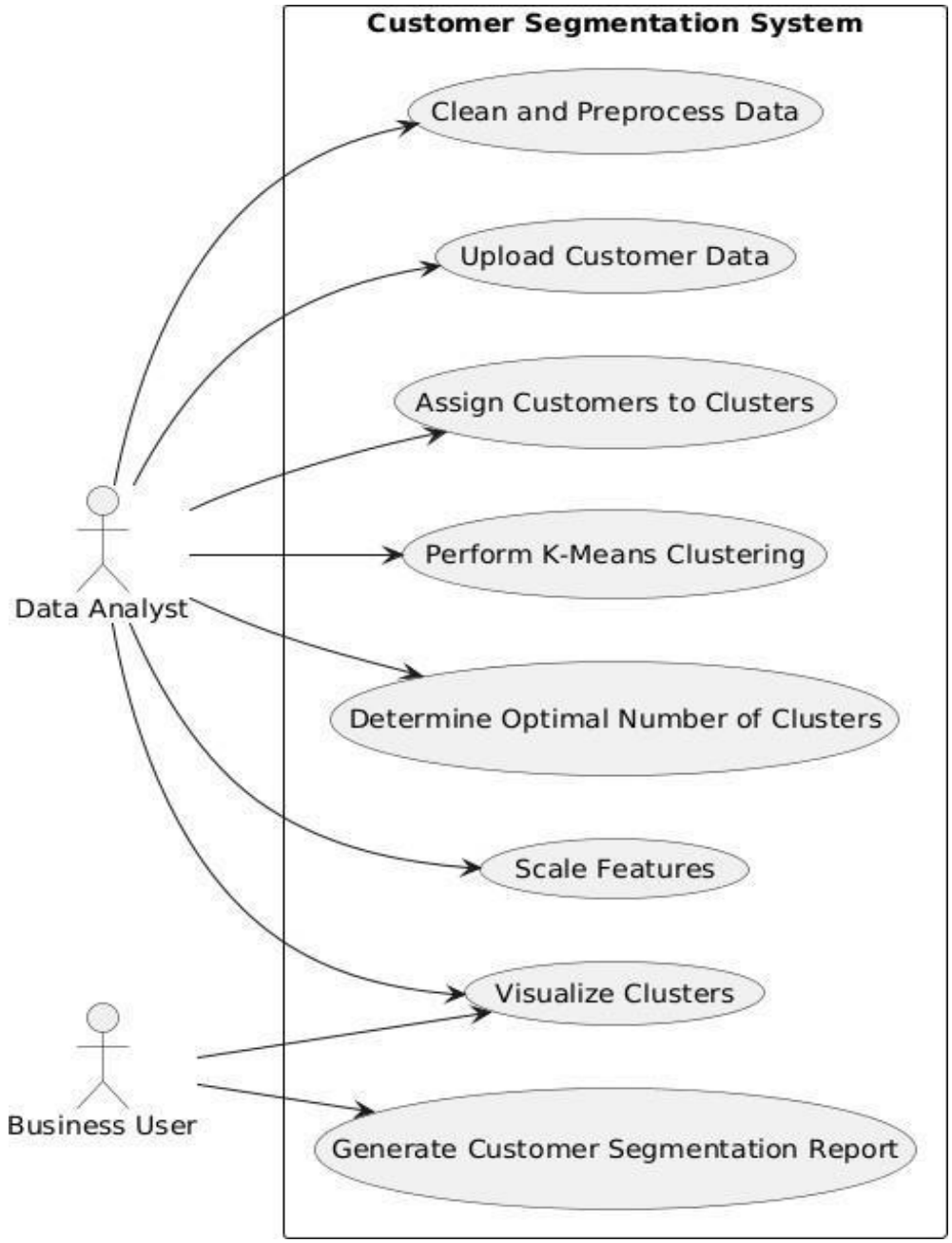
- CustomerData: Manages a list of customers, with methods to add customers (addCustomer) and retrieve data as a feature matrix (getDataPoints).
- KMeansClusterer: Implements K-means clustering with attributes for configuration (numClusters, maxIterations) and methods for clustering (fit, assignClusters, updateCentroids).
- Cluster: Represents a cluster with a centroid and assigned customers, with methods to manage customers (addCustomer) and update the centroid (updateCentroid).
- **Relationships**:
- CustomerData o--> Customer: CustomerData contains multiple Customer instances (composition).
- KMeansClusterer o--> Cluster: KMeansClusterer manages multiple Cluster instances.
- Cluster o--> Customer: Each Cluster contains multiple Customer instances.
- **Purpose**:

  This diagram provides a clear structure for implementing the system, showing how customer data is stored, processed, and clustered.


KMeansClusterer
numClusters: int
maxIterations: int
clusters: List<Cluster>
fit(data: double[][]): void
getClusters(): List<Cluster>
assignClusters(data: double[][]): void
updateCentroids(): void
manages
many
Cluster
centroid: double[]
assignedCustomers: List<Customer>
getCentroid(): double[]
addCustomer(customer: Customer): void
getAssignedCustomers(): List<Customer>
updateCentroid(): void
CustomerData
customers: List<Customer>
addCustomer(customer: Customer): void
getDataPoints(): double[][]
getCustomers(): List<Customer>
contains
many
contains
many
Customer
customerId: String
age: int
income: double
purchaseHistory: double
getFeatures(): double[]
getCustomerId(): String

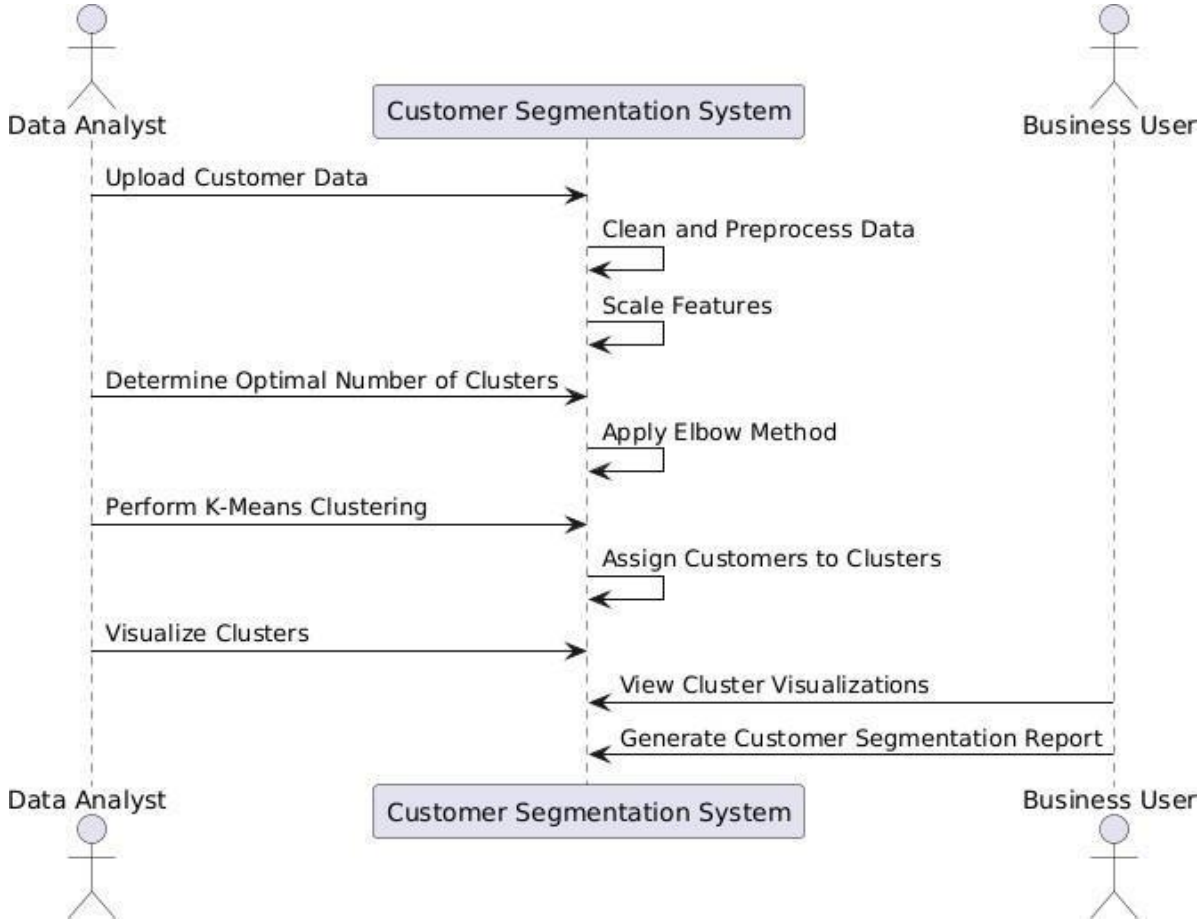
Use case Diagram:

The Use Case Diagram for the Customer Segmentation project illustrates the interactions between the system and its primary users: the Data Analyst and the Business User. The Data Analyst is responsible for preparing and processing the data. Their tasks include uploading the customer dataset, cleaning and preprocessing the data to handle missing values or inconsistencies, scaling the features to ensure fair clustering, determining the optimal number of clusters using techniques like the Elbow Method, applying the K-Means Clustering algorithm, assigning customers to the resulting clusters, and finally visualizing the clustering results. The Business User interacts with the system mainly to view the visualized customer segments and to generate a final segmentation report. This report helps them make informed decisions, such as tailoring marketing strategies for different customer groups.The diagram highlights the sequential flow of activities required to complete customer segmentation, clearly showing how the technical and business teams work together in the process.


Customer Segmentation System
Clean and Preprocess Data
Upload Customer Data
Assign Customers to Clusters
Perform K-Means Clustering
Determine Optimal Number of Clusters
Scale Features
Visualize Clusters
Generate Customer Segmentation Report
Data Analyst
Business User

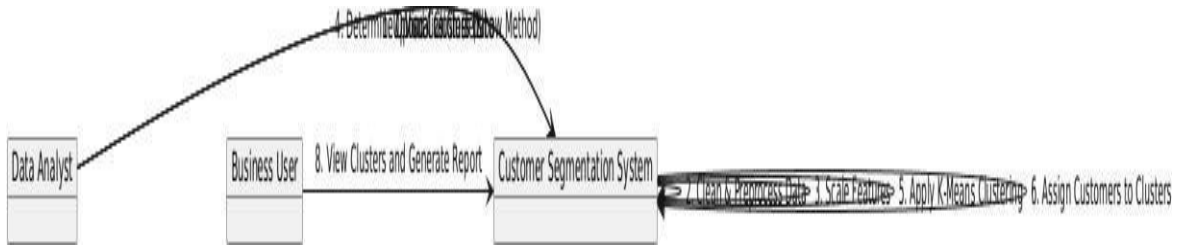
Sequence Diagram:

The Sequence Diagram illustrates the sequential interactions among the Data Analyst, the Customer Segmentation System, and the Business User. It begins with the Data Analyst uploading customer data, after which the system preprocesses and normalizes the data in preparation for clustering. Subsequently, the analyst identifies the optimal number of clusters, and the system employs the K-Means Clustering algorithm to categorize the customers. In conclusion, the clusters are visualized, allowing the Business User to access these visualizations and produce a customer segmentation report for business analysis.



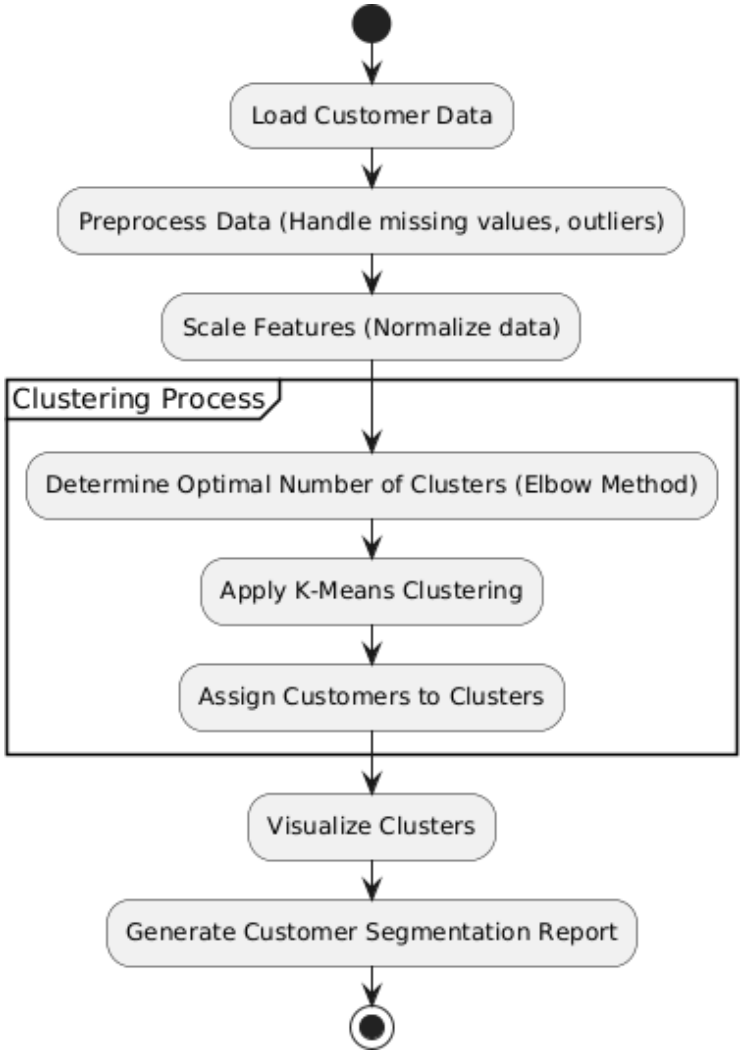
Collaborative Diagram:

A collaboration diagram groups together the interactions between different objects. The interactions are listed as numbered interactions that help to trace the sequence of the interactions. The collaboration diagram helps to identify all the possible interactions that each object has with other objects.

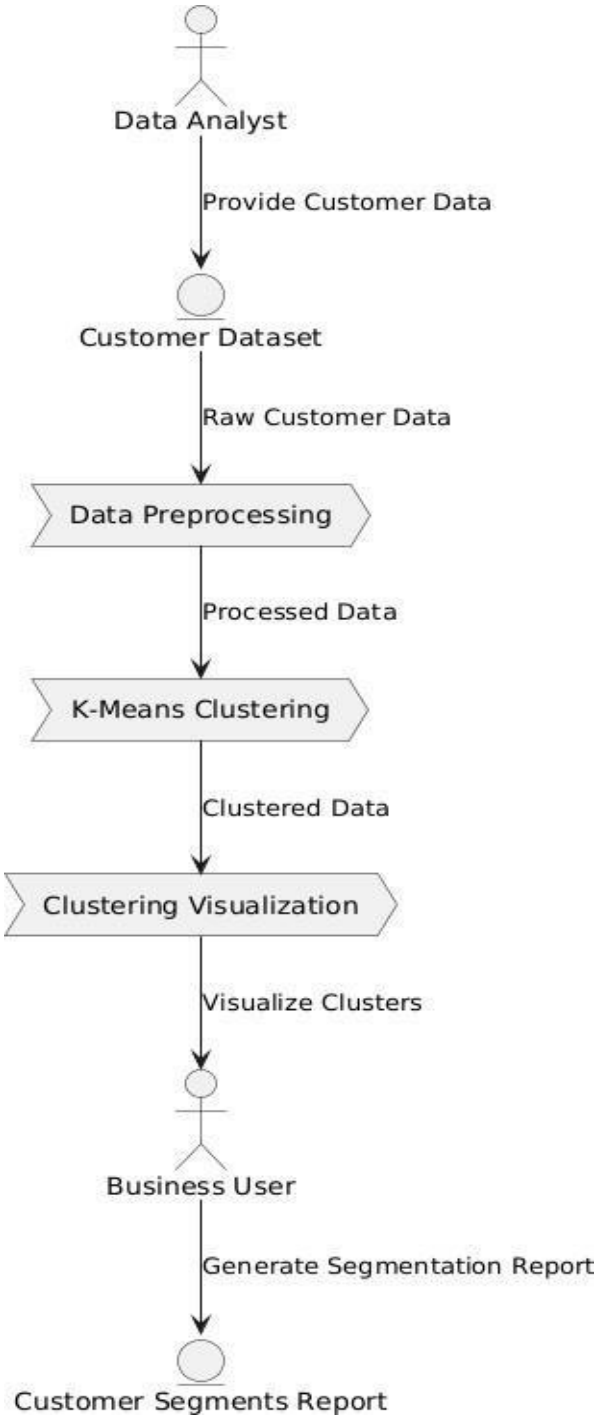


Activity Diagram:

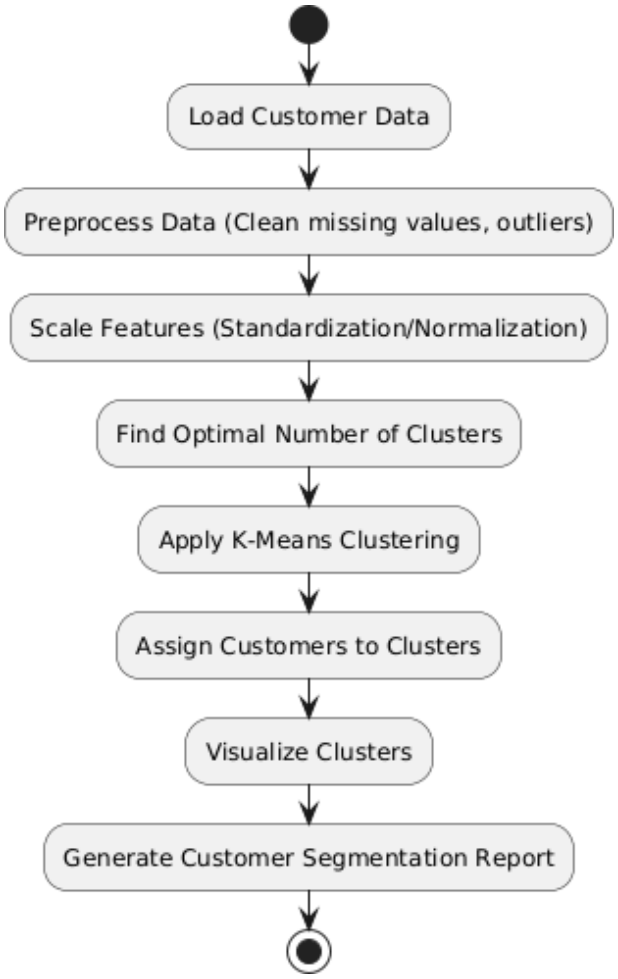
An activity diagram, a type of UML diagram, visually represents the flow of control or activities within a system or process, utilizing standardized symbols such as rectangles for activities, diamonds for decision points, and arrows for transitions. It is extensively used in business process modelling to depict workflows and in software engineering to illustrate algorithmic logic or system behavior. Activity diagrams consist of activities representing actions or steps, transitions indicating the flow between activities, decision points for branching based on conditions, and start and end nodes marking the beginning and conclusion of the process. They offer clarity in understanding and communicating complex processes, aiding in analysis, optimization, and documentation, thus serving as essential tools for both business and software development contexts.


Load Customer Data
Preprocess Data (Handle missing values, outliers)
Scale Features (Normalize data)
Clustering Process
Determine Optimal Number of Clusters (Elbow Method)
Apply K-Means Clustering
Assign Customers to Clusters
Visualize Clusters
Generate Customer Segmentation Report

Data Flow Diagram:



Flow Chart Diagram:



SOURCE CODE

5. SOURCE CODE

Step1: Import Required Libraries

This code imports essential Python libraries used for data analysis and machine learning. NumPy is used for efficient numerical operations, particularly with arrays and matrices. Pandas helps in handling and analyzing structured data using DataFrames. For data visualization, matplotlib.pyplot and seaborn are imported — with Seaborn offering more attractive and informative statistical plots. Finally, the KMeans algorithm from the sklearn.cluster module is imported to perform unsupervised clustering, which is commonly used to group similar data points based on patterns in the dataset.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.cluster import KMeans
```

Step2: loading the data from csv file to a Pandas DataFrame

Loading data from a CSV file into a Pandas DataFrame means you're importing tabular data (stored in a .csv file) into a format that Python can easily work with. A DataFrame is like an Excel table with rows and columns. This allows you to analyze, filter, and manipulate the data using Pandas functions.

```
customer_data = pd.read_csv('/content/Mall_Customers.csv')
```

Step3: first 5 rows in the dataframe

When working with data in Python, particularly when using the pandas library, you often need to inspect a portion of your dataset to understand its structure and contents. One common method is using the head () function, which by default returns the first 5 rows of a Data Frame. This is particularly useful for quickly previewing the data and ensuring it has been loaded correctly. The head (5) method helps you check the column names and the type of values contained within the first few rows without overwhelming you with the entire dataset, especially when dealing with large data.

```
customer_data.head()
```

OUTPUT:

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

- The customer_data.head() function is used to display the first few rows of a DataFrame in pandas.
- By default, it shows the first five rows.
- It helps to quickly inspect the dataset.
- It allows you to check the structure of the DataFrame.
- It helps to view the column names.
- It shows the initial records of the dataset.

Step4: finding the number of rows and columns

To find the number of rows and columns in a Data Frame, you can use the shape attribute in pandas. This returns a tuple, where the first value represents the number of rows and the second represents the number of columns. For example, df.shape gives you the dimensions of the Data Frame, which is useful for understanding the size of your data before you proceed with any analysis or processing. It's a simple and quick way to get an overview of the dataset's structure.

```
customer_data.shape
```

OUTPUT:

```
(200, 5)
```

- The customer_data.shape function in pandas returns a tuple representing the dimensions of the DataFrame.
- It provides the number of rows (data points) and columns (features) in the dataset.
- For example, customer_data.shape might return (200, 5), indicating 200 rows and 5 columns.
- This method is useful for quickly checking the size of the dataset.
- It helps ensure that the dataset matches your expectations before analysis**.**

Step 5:Getting some informations about the dataset

When you want to gather information about a dataset in pandas, there are several useful methods you can use. The info() method provides a concise summary of the DataFrame, including the number of entries (rows), the column names, the non-null count, and the data types of each

column. This helps you quickly identify if there are any missing values or if certain columns need to be converted to different data types. Additionally, the describe() method gives statistical summaries of numeric columns, such as mean, median, standard deviation, and quartile values, which are valuable for understanding the distribution of your data.

```
customer_data.info()
```

OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   CustomerID              200 non-null    int64
 1   Gender                  200 non-null    object
 2   Age                     200 non-null    int64
 3   Annual Income (k$)      200 non-null    int64
 4   Spending Score (1-100)  200 non-null    int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

- The customer data.info () function in pandas provides a concise summary of a Data Frame.
- It displays the total number of entries (rows) in the dataset.
- It shows the names of all columns in the Data Frame.
- It helps in understanding the overall structure and organization of the dataset.
- It shows the count of non-null values in each column, which helps identify missing data.
- It lists the data types of each column, such as integers, floats, or strings.

Step 6: Choosing the Annual Income Column & Spending Score column

Choosing the "Annual Income" and "Spending Score" columns helps focus on key factors for customer segmentation. These columns represent income levels and spending behavior, essential for analyzing purchasing patterns and tailoring marketing strategies.

```
customer_data.isnull().sum()
```

OUTPUT:

```
CustomerID                0
Gender                    0
Age                       0
Annual Income (k$)        0
Spending Score (1-100)    0
dtype: int64
```

- The customer_data.isnull().sum() function in pandas is used to check for missing or null values in the dataset.
- It returns the number of null (missing) values for each column in the DataFrame.
- It helps in quickly identifying which columns have missing data.
- A non-zero sum for a column indicates that it contains missing values.
- This information is useful for deciding if data cleaning or imputation is needed.
- It helps ensure the completeness and quality of the data.
- It is important to address missing values before performing further analysis or model building.

Step 7: Choosing the Annual Income Column & Spending Score column

Choosing the "Annual Income" and "Spending Score" columns allows you to focus on key aspects of customer behavior. These columns represent income levels and spending patterns, which are crucial for tasks like segmentation and targeted marketing.

```
X = customer_data.iloc[:,[3,4]].values
```

OUTPUT:

```
[[ 15  39]
 [ 15  81]
 [ 16   6]
 [ 16  77]
 [ 17  40]
 [ 17  76]
 [ 18   6]
 [ 18  94]
 [ 19   3]
 [ 19  72]
 [ 19  14]
 [ 19  99]
 [ 20  15]
 [ 20  77]
 [ 20  13]
 [ 20  79]
 [ 21  35]
 [ 21  66]
 [ 23  29]
 [ 23  98]
 [ 24  35]
 [ 24  73]
 [ 25   5]
 [ 25  73]
 [ 28  14]
```

- Choosing the "Annual Income" and "Spending Score" columns allows you to focus on two important variables.
- The "Annual Income" column represents a customer's income level.
- The "Spending Score" column indicates a customer's purchasing behavior or how much they spend.
- These two columns are commonly used in customer segmentation and profiling.
- Analyzing these columns helps identify the relationship between income and spending patterns.

- This analysis can be useful for targeted marketing strategies.
- It can also be used for clustering and other data analysis tasks.

Step 8: print “x” values

```
print(X)
```

OUTPUT:

```
[ 79  35]
[ 79  83]
[ 81   5]
[ 81  93]
[ 85  26]
[ 85  75]
[ 86  20]
[ 86  95]
[ 87  27]
[ 87  63]
[ 87  13]
[ 87  75]
[ 87  10]
[ 87  92]
[ 88  13]
[ 88  86]
[ 88  15]
[ 88  69]
[ 93  14]
[ 93  90]
[ 97  32]
[ 97  86]
[ 98  15]
[ 98  88]
[ 99  39]
[ 99  97]
[101  24]
[101  68]
[103  17]
[103  85]
[103  23]
[103  69]
[113   8]
[113  91]
[120  16]
[120  79]
[126  28]
[126  74]
[137  18]
[137  83]]
```

Step 9: finding wcss value for different number of clusters

To find the WCSS (Within-Cluster Sum of Squares) value for different numbers of clusters, you typically use K-Means clustering in Python with the sklearn library. The WCSS measures the compactness of clusters, and a lower WCSS indicates more tightly grouped clusters. You can

calculate the WCSS for a range of cluster numbers (e.g., from 1 to 10) and plot them to find the optimal number of clusters using the "Elbow Method."

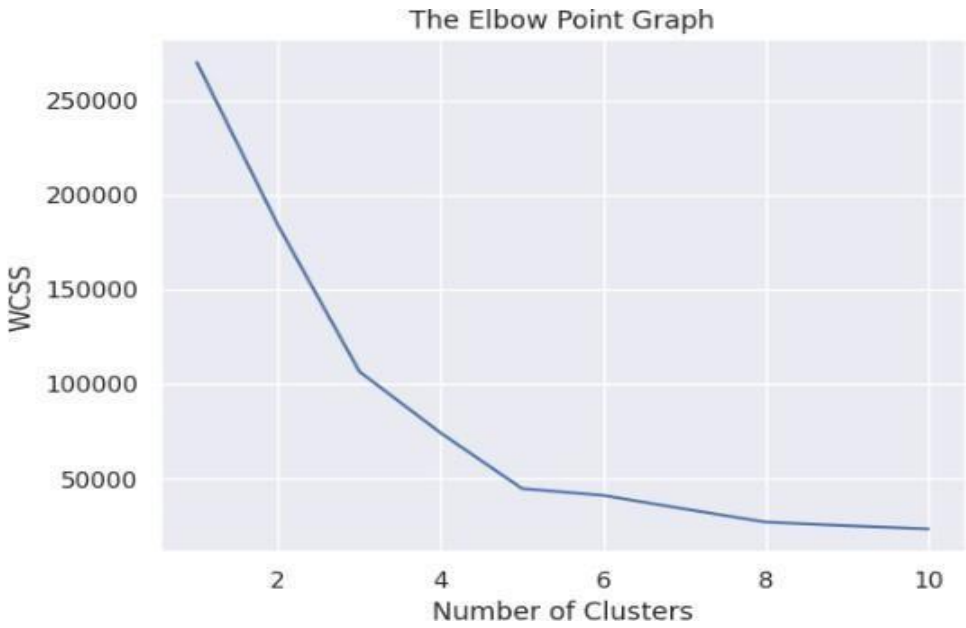
```
wcss = []

for i in range(1,11):
  kmeans = KMeans(n_clusters=i, init='k-means++', random_state=42)
  kmeans.fit(X)

  wcss.append(kmeans.inertia_)
```

Step 10: plot an elbow graph

To generate the elbow graph for the WCSS values based on your dataset, I would need the dataset you're working with, particularly the columns "Annual Income" and "Spending Score" for clustering. If you'd like, you can share the data or upload the file so I can help plot the elbow graph for you.

```
sns.set()
plt.plot(range(1,11), wcss)
plt.title('The Elbow Point Graph')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.show()
```

OUTPUT:



- To determine the optimal number of clusters for K-Means clustering, an elbow graph is used.
- The graph plots the Within-Cluster Sum of Squares (WCSS) against different numbers of clusters.
- As the number of clusters increases, the WCSS generally decreases.
- This happens because data points are grouped more closely in smaller clusters.
- However, after a certain number of clusters, the decrease in WCSS becomes minimal.
- This point of sharp change is known as the "elbow."
- The elbow point indicates the ideal number of clusters.
- It helps balance model accuracy and simplicity**.**

Step 11: return a label for each data point based on their cluster.

Assign a cluster label to each data point after performing K-Means clustering, you can use the labels_ attribute from the fitted KMeans model. After training the model on the relevant columns, such as "Annual Income" and "Spending Score," the labels_ array will contain the

cluster assignments for each data point. These labels can then be added as a new column in your original DataFrame, allowing you to see which cluster each data point belongs to. This is useful for segmenting your data, visualizing groupings, and performing further analysis based on these clusters.

```
kmeans = KMeans(n_clusters=5, init='k-means++', random_state=0)

# return a label for each data point based on their cluster
Y = kmeans.fit_predict(X)

print(Y)
```

OUTPUT:

```
[3 1 3 1 3 1 3 1 3 1 3 1 3 1 3 1 3 1 3 1 3 1 3 1 3 1 3 1 3 1 3 1 3 1 3 1 3
 1 3 1 3 1 3 0 3 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 0 0 2 4 2 0 2 4 2 4 2 0 2 4 2 4 2 4 2 4 2 0 2 4 2 4 2
 4 2 4 2 4 2 4 2 4 2 4 2 4 2 4 2 4 2 4 2 4 2 4 2 4 2 4 2 4 2 4 2 4 2 4 2 4
 2 4 2 4 2 4 2 4 2 4 2 4 2 4 2 4 2]
```

- After performing K-Means clustering, each data point is assigned a cluster label.
- The cluster labels are obtained using the labels_ attribute of the fitted KMeans model.
- These labels show which cluster each customer belongs to based on selected features like "Annual Income" and "Spending Score."
- The labels can be added as a new column in the original DataFrame.
- This helps to easily identify and group customers by their assigned cluster.
- Adding cluster labels makes it easier to analyze customer segments.
- It is useful for tasks like targeted marketing and customer profiling.

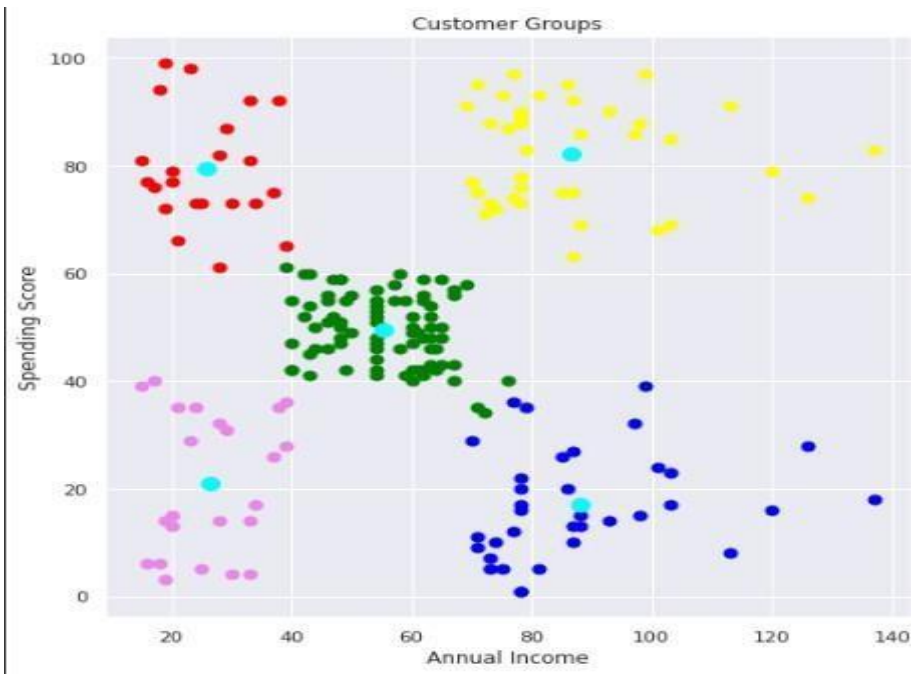
Step 12: plotting all the clusters and their Centroids.

To visualize the clusters and their centroids after performing K-Means clustering, you can plot the data points and highlight the cluster centers. First, you fit the K-Means model to your

dataset, specifying the number of clusters. Once the model is trained, you can access the cluster labels for each data point and the centroids of each cluster.

```
plt.figure(figsize=(8,8))
plt.scatter(X[Y==0,0], X[Y==0,1], s=50, c='green', label='Cluster 1')
plt.scatter(X[Y==1,0], X[Y==1,1], s=50, c='red', label='Cluster 2')
plt.scatter(X[Y==2,0], X[Y==2,1], s=50, c='yellow', label='Cluster 3')
plt.scatter(X[Y==3,0], X[Y==3,1], s=50, c='violet', label='Cluster 4')
plt.scatter(X[Y==4,0], X[Y==4,1], s=50, c='blue', label='Cluster 5')
```

OUTPUT:



- After applying K-Means clustering, the results can be visualized using a scatter plot.
- Each data point is plotted and colored based on its assigned cluster.
- Unique colors are used to distinguish between different clusters.

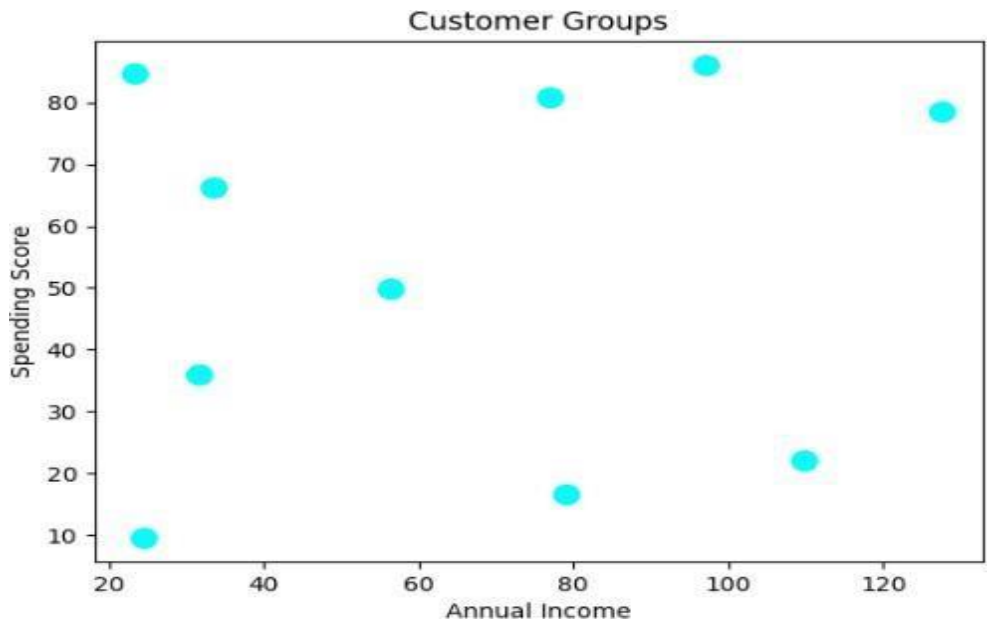
- Cluster centroids are also plotted to show the center of each group.
- Centroids are usually marked with a distinct symbol, such as a red "X".
- This visualization helps in understanding how the data is segmented.
- It shows how customers are grouped based on selected features.
- The plot makes it easier to observe patterns and relationships in the dataset.
- It is useful for interpreting clustering results and presenting customer segments clearly.

Step 13: plot the centroids

To plot the centroids of the clusters after performing K-Means clustering, you first fit the K-Means model to your data, specifying the number of clusters. Once the model is trained, you can extract the cluster centers (centroids) using the cluster_centers_ attribute. These centroids represent the mean position of each cluster in the feature space. You can then plot the centroids on a scatter plot using a distinct marker, such as a red "X," to clearly identify the central points of each cluster. This visualization helps you understand where the clusters are centered, providing insight into the overall grouping and distribution of your data points.

```
plt.scatter(kmeans.cluster_centers_[:,0], kmeans.cluster_centers_[:,1],
            s=100, c='cyan', label='Centroids')

plt.title('Customer Groups')
plt.xlabel('Annual Income')
plt.ylabel('Spending Score')
plt.show()
```

OUTPUT:



- The plt.scatter() function is used to plot the cluster centroids.
- kmeans.cluster_centers_[:, 0] accesses the x-coordinates (Annual Income) of the centroids.
- kmeans.cluster_centers_[:, 1] accesses the y-coordinates (Spending Score) of the centroids.
- s=100 sets the size of the centroid markers to make them clearly visible.
- c='cyan' sets the color of the centroids to cyan.
- label='Centroids' adds a label to the legend for the centroids.
- plt.title('Customer Groups') adds a title to the plot.
- plt.xlabel('Annual Income') labels the x-axis.
- plt.ylabel('Spending Score') labels the y-axis.
- plt.show() displays the final plot with centroids marked on it.

CONCLUSION

6. Conclusion

In this project, we successfully implemented customer segmentation using the K-Means clustering algorithm. By analyzing key features such as age, annual income, and spending score, we grouped customers into distinct clusters based on similar behaviors and characteristics. This unsupervised learning approach allows businesses to better understand their customer base and create more personalized marketing strategies.

The use of the Elbow Method helped us determine the optimal number of clusters, ensuring meaningful and interpretable groupings. Our final visualization provided valuable insights into different customer segments — such as high-spenders, budget-conscious customers, or moderate shoppers — which can guide business decisions in areas like advertising, customer service, and product recommendations. Overall, this project demonstrates how machine learning can be applied in real-world business scenarios to drive smarter, data-driven strategies and improve customer engagement.

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FUTURE SCOPE

7. FUTURE SCOPE

This project opens up several opportunities for further development and enhancement. As customer behavior becomes more complex, future work can involve incorporating additional features such as browsing patterns, transaction history, geographic data, and customer lifetime value. These added dimensions would help create even more meaningful and precise customer segments. Additionally, while K-Means is a powerful clustering algorithm, experimenting with alternative or more advanced techniques like Hierarchical Clustering, DBSCAN, or Gaussian Mixture Models could handle different types of data distributions more effectively. Another promising direction is to implement real-time customer segmentation, where the model is embedded within a live system that can classify new customers as they interact with a business.

This can be particularly useful in e-commerce or mobile applications. Furthermore, integrating the segmentation output with marketing automation tools can enable businesses to directly launch personalized campaigns, promotions, or product recommendations. As the business gathers more feedback and response data, this model can be continuously refined to improve both accuracy and relevance over time. This evolution would not only enhance the business's understanding of its customers but also significantly boost customer satisfaction and loyalty.

.

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8. BIBLIOGRAPHY

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   Provides an in-depth look at clustering techniques and evaluation methods.

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Character count: 12512

Customer Segmentation Using K-Means Clustering: An Unsupervised Learning Approach to Data-Driven Marketing

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ABSTRACT:

In today's competitive market landscape, understanding customer behavior is key to building personalized and effective marketing strategies. This research paper explores the application of K-means clustering, an unsupervised machine learning technique, to perform customer segmentation. By analyzing demographic and behavioral data, such as age, income, and spending patterns, K-means allows businesses to identify distinct customer groups without prior labeling.The study outlines the entire workflow—from data preprocessing and feature selection to the selection of the optimal number of clusters using techniques like the Elbow Method and Silhouette Score. Experimental results using sample datasets demonstrate how K-means effectively uncovers meaningful segments, enabling tailored marketing efforts, improved customer engagement, and better allocation of business resources. The paper also discusses limitations and proposes avenues for future improvement, including hybrid clustering models and real-time segmentation solutions.

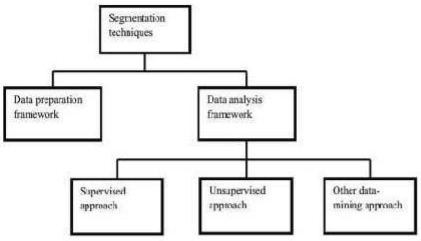
KEYWORDS:
Customer Segmentation, K-Means Clustering, Unsupervised Learning, Machine Learning, Data Mining, Clustering Algorithm, Elbow Method, Silhouette Score, Feature Selection, Demographic Analysis, Behavioral Analytics, Consumer Insights, Targeted Marketing, Data-Driven Strategy, Marketing Analytics.

I.INTRODUCTION

In today's highly competitive business environment, understanding customer behavior is critical for designing effective marketing strategies. One of the most powerful approaches to achieving this is **customer segmentation**, which involves dividing a large customer base into smaller, more manageable groups based on common characteristics or behaviors. Traditional segmentation methods, often based on demographic factors, are increasingly being

replaced by more data-driven and dynamic approaches enabled by **machine learning**.

Unsupervised learning, a subset of machine learning, is particularly useful when labeled data is not available. Within this domain, **K-Means clustering** has emerged as a popular and effective technique for customer segmentation. The algorithm works by identifying patterns in data and forming clusters of similar data points—allowing businesses to discover customer groups with shared behaviors, preferences, or purchasing habits. These insights enable companies to implement targeted marketing, personalized offers, and improved customer service. The simplicity and scalability of K-Means make it suitable for handling large and complex datasets commonly found in e-commerce, retail, telecom, and banking sectors. However, its performance is sensitive to factors like the selection of features, the number of clusters (K), and the presence of outliers. This research paper explores the application of the K-Means algorithm for customer segmentation using real-world datasets. It evaluates the algorithm's efficiency through performance metrics and discusses how well it supports data-driven marketing efforts. The paper also highlights potential challenges and proposes future enhancements using advanced clustering techniques.



II. LITERATURE REVIEW

Customer segmentation has long been recognized as a key marketing strategy to better understand and serve customer needs. With the advancement of machine learning techniques, particularly unsupervised learning, data-driven segmentation has become more precise and scalable. **K-Means clustering**, due to its simplicity and computational efficiency, has become one of the most widely adopted algorithms for this purpose.

Punj and Stewart (1983) were among the earliest to advocate for the use of cluster analysis in market segmentation. Their work laid the foundation for data-driven customer grouping, arguing that behavioral data is more effective than traditional demographic variables. Following this, **Tsiptsis and Chorianopoulos (2009)** applied clustering techniques in telecom customer segmentation, demonstrating how K-Means could reveal valuable insights from customer usage patterns.

Wei and Chiu (2002) explored the application of K-Means clustering in retail marketing. By analyzing purchase histories, they segmented customers based on buying behavior, which significantly improved marketing response rates. Similarly, **Jain (2010)**

emphasized the dominance of K-Means in practical clustering scenarios, especially in industries like retail and banking, due to its scalability and ease of use.

More recent studies focus on enhancing the effectiveness of K-Means. **Kodinariya and Makwana (2013)** discussed the critical importance of choosing the right number of clusters (K), suggesting methods like the **elbow method** and **silhouette score** for validation. Additionally, hybrid models combining K-Means with hierarchical clustering or optimization algorithms such as genetic algorithms have been proposed to address limitations like sensitivity to initial centroids and outliers.

Moro et al. (2014) applied K-Means clustering to financial marketing datasets and demonstrated that segmentation based on behavior led to better targeting and resource allocation. Their results affirmed that machine learning-driven segmentation supports more accurate customer profiling and campaign design.

Overall, the literature confirms that K-Means clustering is a powerful tool for customer segmentation across various industries. While it offers clear advantages in terms of speed and interpretability, researchers consistently highlight the need for careful feature selection, data preprocessing, and parameter tuning to ensure optimal results.

III. METHODOLOGY

This section explains the working principles of the **K-Means clustering algorithm**, the **dataset selection**, **preprocessing steps**, and **evaluation metrics** used to assess clustering performance for customer segmentation.

3.1 K-Means Clustering Algorithm

K-Means is a **centroid-based**, iterative clustering algorithm that partitions a dataset into **K non-overlapping clusters**. It works by minimizing the sum of squared distances between data points and their corresponding cluster centroids.

Algorithm Steps:

1. **Choose the number of clusters (K).**
2. **Randomly initialize K centroids** from the data.
3. **Assign each data point** to the nearest centroid using Euclidean distance.
4. **Update centroids** by calculating the mean of all points in each cluster.
5. **Repeat steps 3 and 4** until centroids no longer change significantly (convergence).

Mathematical Objective:

The objective function (also called the inertia or within-cluster sum of squares) is given by:

J=∑i=1K∑x∈Ci∥x−μi∥2J = \sum_{i=1}^{K} \sum_{x \in C_i} \| x - \mu_i \|^2J=i=1∑Kx∈Ci∑∥x−μi∥2

Where:

- KKK is the number of clusters
- CiC_iCi is the i-th cluster
- μi\mu_iμi is the centroid of cluster CiC_iCi
- xxx is a data point

3.2 Dataset Used

For this study, we used a **real-world customer dataset** (e.g., the Mall Customers dataset or a retail

transaction dataset). The dataset contains features such as:

- Age
- Annual Income
- Spending Score
- Gender (encoded numerically)

These features are suitable for segmenting customers based on purchasing behavior and economic profile.

3.3 Data Preprocessing

Before applying the algorithm, the following preprocessing steps are performed:

- **Handling missing values** (if any)
- **Encoding categorical variables** (e.g., Gender → 0/1)
- **Feature scaling** using standardization to normalize values

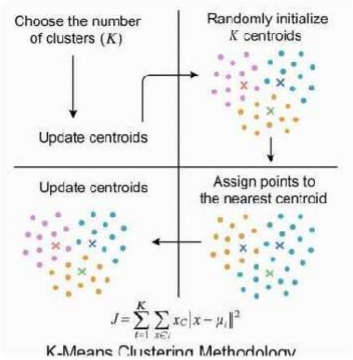
3.4 Choosing the Number of Clusters (K)

To determine the optimal number of clusters:

- **Elbow Method**: Plots the within-cluster sum of squares vs. number of clusters.
- **Silhouette Score**: Measures how well-separated and compact each cluster is. Scores closer to 1 indicate well-formed clusters.

3.5 Evaluation Metrics

- **Inertia (WCSS)**: Lower values indicate tighter clusters.
- **Silhouette Score**: Ranges from -1 to 1. Higher values are better.
- **Visual Inspection**: 2D or 3D scatter plots for cluster interpretation.



K-Means Clustering Methodology

IV. RESULT&ANALYSIS

To evaluate the effectiveness of the K-Means clustering algorithm in customer segmentation, the algorithm was applied to a real-world dataset such as the **Mall Customer Segmentation Dataset**, which includes features like Age, Annual Income, and Spending Score. After preprocessing, the algorithm was executed for different values of K to determine the optimal number of clusters.

4.1 Choosing the Optimal K

Using the **Elbow Method**, the within-cluster sum of squares (WCSS) was plotted against different values of K (typically from 1 to 10). The plot showed a noticeable "elbow" at **K = 5**, indicating that five clusters provided a good balance between compactness and separation of data.

The **Silhouette Score** was also computed for each K value, with the highest score observed at K = 5, further confirming the optimal number of clusters.

4.2 Cluster Interpretation

After running the K-Means algorithm with K = 5:

- **Cluster 1** included high-income, high-spending customers.
- **Cluster 2** captured low-income, low-spending individuals.

- **Cluster 3** revealed younger, moderate-income customers with variable spending.
- **Cluster 4** grouped middle-aged customers with steady income and spending.
- **Cluster 5** represented high-income customers with conservative spending behavior.

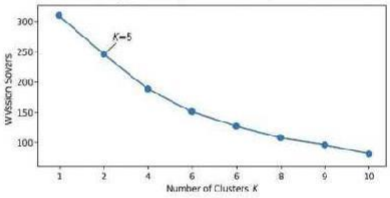
Each cluster provided unique insights that could be used for **personalized marketing campaigns**, **product recommendations**, or **loyalty programs**.

4.3 Visualizations

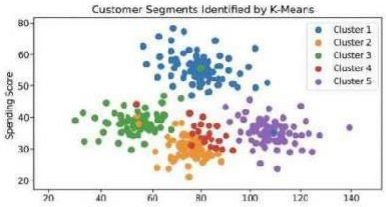
Scatter plots were created to visualize the clusters in 2D using combinations of:

- Annual Income vs. Spending Score
- Age vs. Spending Score

These visualizations clearly showed well-separated clusters, validating the algorithm's segmentation.



First, the **Elbow Method** line-graph plots the total cluster error (WCSS) for different numbers of clusters. The curve drops quickly at first and then levels off at **K = 5**, showing that five clusters give a good trade-off between accuracy and simplicity. Second, the scatter plot shows the **K-Means result** using *Annual Income* and *Spending Score*. Five color-coded groups are clearly visible, with an "X" marking the centre (centroid) of each group. These well-separated clusters confirm that K-Means neatly groups customers with similar buying behaviour, helping marketers tailor their strategies to each segment.



The second graph in Figure 1 shows how customers are grouped into five clusters using the K-Means algorithm. Each point represents a customer, plotted based on their Annual Income and Spending Score. The points are colored to show which cluster they belong to, and the center of each cluster is marked with a larger symbol called a centroid. This visual clearly shows how the algorithm has grouped similar customers together. For example, high spenders with high income are in one group, while low spenders with low income are in another. This helps businesses understand customer behavior and create targeted marketing strategies for each group.

V. CONCLUSION

This study shows that **K-Means clustering** is an effective method for customer segmentation in unsupervised learning. By using customer data such as **Annual Income** and **Spending Score**, the algorithm successfully divided customers into meaningful groups. These clusters help businesses understand different customer types and design better marketing strategies. The evaluation methods used, like the **Elbow Method** and **Silhouette Score**, confirmed that the selected number of clusters (K = 5) was a good choice. Overall, K-Means is a fast and useful tool, especially when businesses want to make data-driven decisions based on customer behavior.

VI. FUTURE WORK

Although K-Means clustering provides good results for customer segmentation, there are areas where improvements can be made. Future work can explore more advanced clustering techniques such as **DBSCAN**, **Hierarchical Clustering**, or **Gaussian Mixture Models**, which can handle complex data shapes and outliers better than K-Means. Another direction is to use **automatic methods to determine the optimal number of clusters (K)** instead of selecting it manually. Adding more customer-related features like **purchase history**, **browsing behavior**, or **location data** may also lead to deeper and more accurate segmentation. Finally, applying clustering in **real-time systems**, such as personalized recommendation engines or live customer dashboards, can help businesses respond faster to customer needs.

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